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Unit 6: Data Science and Data Gathering

Complete Exam Question Bank — MCQs, Short Questions & Long Questions

Section A: Multiple Choice Questions

Each question has one correct answer. The correct answer is in bold.

Topic: Data and Data Types (Qualitative vs Quantitative)

1. What is data?

- a) Processed information ready for use
 - b) **Raw facts gathered about things around us that can be processed to generate useful information**
 - c) A collection of numbers only
 - d) A list of observed events already interpreted
-

2. Which of the following best describes qualitative data?

- a) Data expressed in numbers and used for mathematical calculations
 - b) Data that can be measured using instruments
 - c) **Data that describes qualities or characteristics using words, labels, or categories**
 - d) Data that can be counted or enumerated
-

3. Which of the following is an example of qualitative data?

- a) Temperature readings in degrees Celsius
 - b) Number of students in a class
 - c) **Favourite ice cream flavours**
 - d) Test scores out of 100
-

4. Which of the following is an example of quantitative data?

- a) Names of students in a class
 - b) Types of fruit
 - c) Colours of cars in a parking lot
 - d) **Heights and weights of students**
-

5. Which characteristic describes qualitative data?

- a) Numerical and measurable
 - b) Countable and arithmetical
 - c) **Non-numeric, descriptive, and categorical**
 - d) Expressed in units like kilograms or metres
-

6. Which of the following questions would generate quantitative data?

- a) "What is your favourite colour?"
 - b) "Which subject do you enjoy the most?"
 - c) **"How many hours do you study per day?"**
 - d) "What type of music do you prefer?"
-

7. Which of the following questions would generate qualitative data?

- a) "How tall are you?"
 - b) "How many books have you read?"
 - c) "What is your test score?"
 - d) **"What is your favourite school activity?"**
-

8. Multiplying the unit price of a fruit by its weight to find the total price is an example of which characteristic of quantitative data?

- a) Numerical
 - b) Measurable
 - c) Countable
 - d) **Arithmetical**
-

9. The names of students in a class — Ali, Badar, Qasim — are an example of which type of data?

- a) Quantitative — numerical data
 - b) Quantitative — measurable data
 - c) **Qualitative — non-numeric data**
 - d) Quantitative — countable data
-

10. The number of cars in a car park is an example of which characteristic of quantitative data?

- a) Arithmetical
 - b) Measurable
 - c) Numerical
 - d) **Countable**
-

11. Using a thermometer to measure temperature is an example of which characteristic of quantitative data?

- a) Countable
 - b) Arithmetical
 - c) Numerical
 - d) **Measurable**
-

12. Which of the following pairs correctly classifies the data type?

- a) Number of students = qualitative; student names = quantitative
 - b) Test scores = qualitative; student names = quantitative
 - c) **Student names = qualitative; test scores = quantitative**
 - d) Favourite subjects = quantitative; heights = qualitative
-

Topic: Organising and Analysing Data

13. Why is organising data important before analysis?

- a) It makes data look more colourful
- b) It reduces the amount of data collected
- c) **It reduces errors, saves time, and improves clarity**
- d) It automatically generates conclusions

14. Which of the following is the best tool for presenting students' scores across multiple subjects clearly?

- a) A paragraph of text
 - b) A random list
 - c) An audio recording
 - d) **A data table**
-

15. Charts in data analysis are used primarily to:

- a) Store large amounts of data permanently
 - b) Replace data tables entirely
 - c) **Identify patterns, trends, and outliers in datasets visually**
 - d) Automatically analyse data without human input
-

16. Which of the following is NOT a common type of chart?

- a) Bar chart
 - b) Line chart
 - c) Pie chart
 - d) **Flow chart**
-

17. Which of the following is NOT a common type of graph?

- a) Line graph
 - b) Bar graph
 - c) Scatter plot
 - d) **Pie diagram**
-

18. A scatter plot is best used to show:

- a) The proportion of a whole divided into categories
 - b) Trends over a period of time
 - c) **The relationship between two different variables**
 - d) A comparison of quantities using rectangular bars
-

19. A pie chart is most suitable for showing:

- a) Trends over time
 - b) The relationship between two variables
 - c) Distribution of values in a dataset
 - d) **The proportion of different categories as parts of a whole**
-

20. A line graph is most suitable for showing:

- a) Proportion of categories in a whole
 - b) Relationship between two unrelated variables
 - c) Comparison of individual items side by side
 - d) **Trends and changes in data over a period of time**
-

Topic: Data Collection Methods and Tools

21. What is data collection?

- a) Storing processed data in a database
 - b) Deleting old and unnecessary data
 - c) Analysing data to find patterns
 - d) **The process of gathering information to answer questions or make decisions**
-

22. Which data collection method gathers information from people by asking them questions, either on paper, over the phone, or online?

- a) Observation
 - b) Interview
 - c) Online data sources
 - d) **Survey**
-

23. Which data collection method involves written forms with a set of questions that people fill out, often with multiple-choice answers?

- a) Survey
- b) Interview
- c) **Questionnaire**
- d) Observation

24. Which data collection method involves talking to individuals one-on-one to gather detailed information?

- a) Survey
 - b) Questionnaire
 - c) **Interview**
 - d) Online data source
-

25. A teacher watches how students behave during a group project to understand their teamwork. Which data collection method is this?

- a) Survey
 - b) Interview
 - c) Questionnaire
 - d) **Observation**
-

26. Using websites, databases, and digital tools to find and collect information is known as:

- a) Survey
 - b) Observation
 - c) Interview
 - d) **Online data sources**
-

27. Which of the following is a free online tool from Google used to create surveys?

- a) SurveyMonkey
 - b) Microsoft Forms
 - c) **Google Forms**
 - d) Tableau
-

28. Which of the following is a best practice in survey design?

- a) Use long, complex questions to gather more information
- b) Include as many questions as possible
- c) Only send the survey to one group of people
- d) **Keep it short, simple, and test the survey before sending it out**

29. Which of the following platforms is commonly used for creating detailed professional surveys?

- a) Google Docs
 - b) Microsoft Excel
 - c) **SurveyMonkey**
 - d) Tableau
-

30. To collect the most detailed and in-depth information from an individual, the most appropriate method would be:

- a) Survey
 - b) Online data sources
 - c) Observation
 - d) **Interview**
-

Topic: Structured and Unstructured Data

31. With respect to storage and processing, data is divided into which two types?

- a) Qualitative and quantitative
 - b) Numeric and non-numeric
 - c) **Structured and unstructured**
 - d) Primary and secondary
-

32. Which of the following best describes structured data?

- a) Data that is free-form and does not follow any format
 - b) Data found in emails and social media posts
 - c) **Data that is organised and formatted to be easily searchable and analysable**
 - d) Data represented as images and videos
-

33. Which of the following is an example of structured data?

- a) A social media post
- b) An email
- c) A video file
- d) **Data in a spreadsheet or traditional database**

34. Which of the following is an example of unstructured data?

- a) A table of student marks
 - b) A database of product prices
 - c) **A social media post or video**
 - d) A spreadsheet of employee details
-

35. A student database with columns for Student ID, Name, Class, Date of Birth, and Fee Status is an example of:

- a) Unstructured data
 - b) Qualitative data only
 - c) Raw data
 - d) **Structured data**
-

36. Unstructured data is described as:

- a) Organised in rows and columns in a spreadsheet
 - b) Easily searchable using standard database tools
 - c) Expressed only in numbers
 - d) **Free-form data that does not fit into a specific format**
-

Topic: Data Visualisation

37. What is data visualisation?

- a) The process of collecting raw data from various sources
 - b) The process of storing data in a database
 - c) **The process of turning numbers and information into pictures such as charts and graphs**
 - d) The process of deleting unnecessary data
-

38. What is the primary purpose of data visualisation?

- a) To generate random numbers
- b) To convert text into structured data
- c) To hide complex relationships in data

- d) **To make data easier to understand by representing it visually**
-

39. Which of the following is a data visualisation tool specifically designed for creating detailed and interactive visualisations?

- a) Microsoft Word
 - b) Google Docs
 - c) Python
 - d) **Tableau**
-

40. Which of the following tools allows users to create and share visualisations online in a way similar to Microsoft Excel?

- a) Tableau
 - b) Power BI
 - c) **Google Sheets**
 - d) SQL
-

41. Microsoft Power BI is best described as:

- a) A programming language for data analysis
 - b) A spreadsheet tool for basic calculations
 - c) A database management system
 - d) **A tool that allows users to create a wide variety of visualisations including charts, graphs, and maps**
-

42. Which visualisation tool is most suitable for students who already know spreadsheet software and want to create basic charts quickly?

- a) Tableau
 - b) Power BI
 - c) Python
 - d) **Microsoft Excel**
-

43. A bar chart is most useful for:

- a) Showing trends over time

- b) Showing proportions as parts of a whole
 - c) Showing relationships between two variables
 - d) **Comparing quantities across different categories**
-

44. A histogram is used to show:

- a) The proportion of categories in a dataset
 - b) Trends in data over time
 - c) Relationships between two different variables
 - d) **The frequency distribution of a dataset**
-

Topic: Data Pre-Processing

45. What is data pre-processing?

- a) The final step in data analysis where conclusions are drawn
 - b) The process of collecting data from surveys and questionnaires
 - c) The process of visualising data using charts and graphs
 - d) **The first and most important step of preparing data for analysis by cleaning and organising it**
-

46. Which analogy best describes data pre-processing?

- a) Writing a report after completing research
 - b) Presenting a finished dish at a restaurant
 - c) Eating a meal at a restaurant
 - d) **Preparing ingredients before cooking — washing, chopping, and measuring everything**
-

47. What does evaluating data quality involve?

- a) Creating visualisations from the data
 - b) Collecting more data from additional sources
 - c) Presenting data in tables and charts
 - d) **Checking if data is accurate, complete, consistent, and reliable**
-

48. In a list of student test scores, if one student's score is recorded as 105 out of 100, this is an example of:

- a) An outlier
 - b) A bias
 - c) Missing data
 - d) **An error**
-

49. In a list of test scores where most students scored between 50 and 80, but one student scored 5, the score of 5 is an example of:

- a) An error
 - b) A bias
 - c) Missing data
 - d) **An outlier**
-

50. If a survey only includes responses from students in one school but is used to represent opinions of all students in a city, this is an example of:

- a) An error
 - b) An outlier
 - c) Missing data
 - d) **A bias**
-

51. Data validation involves:

- a) Deleting all data that contains errors
 - b) Creating visualisations of the data
 - c) Collecting additional data to replace missing values
 - d) **Checking data for completeness and accuracy**
-

52. Ensuring that every student has a test score recorded in a dataset is an example of:

- a) Validating data accuracy
- b) Removing outliers
- c) Handling missing data
- d) **Validating data completeness**

53. Verifying that all test scores in a dataset are between 0 and 100 is an example of:

- a) Validating data completeness
 - b) Removing outliers
 - c) Handling missing data
 - d) **Validating data accuracy**
-

54. When a student's test score is missing and you fill in the gap using the class average, this is an example of:

- a) Data validation
 - b) Removing outliers
 - c) Error correction
 - d) **Handling missing data**
-

55. Data cleaning involves which of the following?

- a) Only collecting new data to replace old data
 - b) Creating charts and graphs from existing data
 - c) Storing data in a structured format
 - d) **Removing errors, handling missing data, and dealing with outliers**
-

56. Which of the following is NOT a data pre-processing technique?

- a) Identifying errors and outliers
 - b) Validating data completeness
 - c) Handling missing data
 - d) **Creating a bar chart from the data**
-

Topic: Data Analysis Techniques

57. What is the purpose of data analysis?

- a) To collect data from multiple sources
- b) To store data in a structured format
- c) To visualise data using charts

- d) **To examine data to discover useful information, draw conclusions, and support decision-making**
-

58. Quantitative analysis deals with:

- a) Text, images, and sounds
 - b) Descriptions and observations
 - c) Experiences and meanings
 - d) **Numbers and measurable data**
-

59. Qualitative analysis deals with:

- a) Numbers and statistical calculations
 - b) Measurable and countable data
 - c) Mathematical formulas and equations
 - d) **Non-numeric data such as text, images, and sounds**
-

60. Which type of analysis helps us understand patterns, relationships, and trends in numeric data?

- a) Qualitative analysis
 - b) Descriptive analysis
 - c) **Quantitative analysis**
 - d) Categorical analysis
-

61. Which type of analysis helps us understand meanings, concepts, and experiences?

- a) Quantitative analysis
 - b) Statistical analysis
 - c) Numerical analysis
 - d) **Qualitative analysis**
-

Topic: Cloud Storage and Data Backups

62. What is cloud storage?

- a) Storing data on a USB flash drive

- b) Storing data on a physical hard drive at home
 - c) **Saving files on the internet for access from any device**
 - d) Storing data on a DVD or CD
-

63. What is remote access in the context of data management?

- a) Accessing files stored only on the local computer
 - b) Printing documents from a wireless printer nearby
 - c) Connecting two computers using a USB cable
 - d) **The ability to connect to and use a computer or network from a distant location**
-

64. What is a data backup?

- a) The original copy of a file stored on your computer
 - b) A method of compressing data to save space
 - c) A process of encrypting data for security
 - d) **A copy of important data stored separately from the original to protect against data loss**
-

65. Why are data backups important?

- a) They make files load faster
 - b) They increase the storage capacity of a computer
 - c) They allow multiple users to edit the same file at once
 - d) **They allow you to recover data if it is lost due to accidental deletion, hardware failure, or a virus**
-

66. A student saves a school project on their computer and also uploads a copy to Google Drive. The Google Drive copy is an example of:

- a) Data compression
 - b) Data validation
 - c) **A data backup**
 - d) Data pre-processing
-

67. Which of the following is a benefit of using collaborative cloud tools in data management?

- a) They reduce the need for any data analysis
 - b) They eliminate the possibility of data errors
 - c) They replace the need for any physical devices
 - d) **They allow multiple people to work on the same project simultaneously from different locations**
-

68. Version control in collaborative tools like Google Docs means:

- a) The document is automatically deleted after a set time
 - b) Only one person can edit the document at a time
 - c) The document is converted to a different file format
 - d) **Every change is saved automatically, allowing users to go back to previous versions**
-

69. Which of the following is an example of a cloud storage service?

- a) Microsoft Excel
 - b) USB flash drive
 - c) **Google Drive**
 - d) SQL Server
-

70. Which of the following correctly describes a benefit of collaborative tools?

- a) They slow down the data collection process
 - b) They prevent multiple users from accessing the same file
 - c) They require all team members to be in the same physical location
 - d) **They enhance productivity by allowing multiple users to work on different sections of a project simultaneously**
-

Topic: Introduction to Data Science

71. Data science is best described as:

- a) A tool for creating spreadsheets and charts
- b) A method of storing large amounts of structured data
- c) A programming language used for web development

- d) **A field that combines computer skills, mathematics, and business knowledge to turn raw data into useful insights**
-

72. Which analogy is used in the chapter to describe data science?

- a) A chef preparing ingredients for a meal
 - b) A librarian organising books on a shelf
 - c) A teacher grading exam papers
 - d) **A detective solving problems using data instead of solving crimes**
-

73. Which of the following fields is NOT part of the interdisciplinary nature of data science?

- a) Computer Science
 - b) Mathematics and Statistics
 - c) Business Knowledge
 - d) **Literature and History**
-

74. Why is knowing about data science important in daily life?

- a) It allows you to write programs in all languages
 - b) It helps you build websites and mobile applications
 - c) It is only useful for professional data scientists
 - d) **It helps us make better decisions, understand patterns, and solve problems more effectively**
-

75. Which of the following is an example of data science being used in sports?

- a) Designing uniforms using computer graphics
 - b) Broadcasting matches on television
 - c) **Using data to create better strategies and enhance team performance**
 - d) Building larger stadiums using architectural software
-

76. Which of the following is an example of data science being used in healthcare?

- a) Ordering medical equipment online
- b) Scheduling doctor appointments using a calendar app
- c) **Predicting health trends and providing better patient care**

- d) Designing hospital buildings using CAD software
-

77. The role of Computer Science in data science is to:

- a) Apply insights to real-life business problems
 - b) Analyse data and find statistical patterns
 - c) Create reports and visualisations for clients
 - d) **Handle and organise data**
-

78. The role of Mathematics and Statistics in data science is to:

- a) Handle and organise data using computers
 - b) Apply insights to real-life business problems
 - c) Create visualisations for presentations
 - d) **Analyse data and find patterns**
-

79. The role of Business Knowledge in data science is to:

- a) Write programs to collect data
 - b) Perform statistical calculations on datasets
 - c) Design databases for data storage
 - d) **Apply data insights to real-life problems and make informed decisions**
-

Topic: Data Science Tools

80. Which data science tool is a widely-used spreadsheet program that helps organise and analyse data?

- a) Python
 - b) R
 - c) SQL
 - d) **Microsoft Excel**
-

81. Which data science tool is a popular programming language used for analysing data and building data models?

- a) Excel

- b) SQL
 - c) Tableau
 - d) **Python**
-

82. Which Python library is used for data manipulation and analysis?

- a) Matplotlib
 - b) NumPy
 - c) **Pandas**
 - d) SciPy
-

83. Which Python library is used for creating graphs and charts?

- a) Pandas
 - b) NumPy
 - c) **Matplotlib**
 - d) SciPy
-

84. Which data science tool is specifically designed for statistical analysis and data visualisation, and is particularly good at handling complex data?

- a) Python
 - b) Excel
 - c) SQL
 - d) **R**
-

85. What does SQL stand for?

- a) Simple Question Language
 - b) System Query Logic
 - c) Software Query Library
 - d) **Structured Query Language**
-

86. SQL is primarily used for:

- a) Creating visualisations and dashboards

- b) Building machine learning models
 - c) Writing statistical analysis reports
 - d) **Managing and querying databases**
-

87. If you want to find all students who scored above 90 in a large database, which tool would you use?

- a) Excel
 - b) Python
 - c) **SQL**
 - d) Tableau
-

88. Which analogy is used in the chapter to describe the role of SQL?

- a) A chef preparing ingredients
 - b) A detective solving a crime
 - c) A GPS finding the best route
 - d) **A librarian finding all books by a particular author from a huge collection**
-

89. A student wants to analyse survey results and then create a chart showing the findings. Which set of Python tools would they use?

- a) SQL and Tableau
 - b) R and Excel
 - c) **Pandas and Matplotlib**
 - d) Power BI and Google Sheets
-

90. Which data science tool would a scientist use to analyse a dataset from a scientific experiment and create detailed statistical plots?

- a) SQL
 - b) Excel
 - c) **R**
 - d) Tableau
-

Topic: Big Data and Data Storage

91. Big data refers to:

- a) Data stored only in spreadsheets
 - b) Small datasets that are easy to analyse
 - c) Only quantitative data collected from surveys
 - d) **Extremely large and complex datasets that cannot be processed using traditional tools**
-

92. Which of the following is NOT a domain where big data is commonly applied?

- a) Retail
 - b) Healthcare
 - c) Finance
 - d) **Creative writing**
-

93. Which type of data storage is best suited for organised, easily searchable data in rows and columns?

- a) Cloud storage
 - b) NoSQL database
 - c) Data warehouse
 - d) **Spreadsheet or traditional relational database**
-

94. A data warehouse is best described as:

- a) A physical room where computers are stored
 - b) A tool for creating visualisations
 - c) **A large storage system used to store and manage large volumes of structured data for analysis**
 - d) A type of unstructured data storage
-

95. NoSQL databases are best suited for:

- a) Storing data only in tables with fixed columns
- b) Running SQL queries on structured data
- c) Creating charts and graphs from data

- d) **Storing large volumes of unstructured or semi-structured data that does not fit neatly into tables**
-

96. Which of the following is an example of data science being applied in retail?

- a) Designing store layouts using architecture software
 - b) Training store employees in customer service
 - c) **Analysing customer purchase patterns to recommend products**
 - d) Printing price tags using label software
-

97. Predictive modelling in data science means:

- a) Creating charts and graphs from historical data
 - b) Cleaning and organising data before analysis
 - c) Collecting data from surveys and questionnaires
 - d) **Using historical data to make predictions about future outcomes**
-

98. Graph analytics in data science is used to:

- a) Create bar charts and pie charts for presentations
 - b) Store data in graph-based NoSQL databases
 - c) Write SQL queries to extract data from databases
 - d) **Analyse relationships and connections between data points**
-

99. Data Analytics is described in the chapter as:

- a) A method of collecting data from multiple sources
 - b) A tool for creating interactive visualisations
 - c) A programming language for data science
 - d) **A branch of data science focused on analysing data to understand it better and support decision-making**
-

100. Which of the following is the correct order of steps in a typical data science process?

- a) Data Analysis → Data Collection → Data Cleaning → Understanding the Problem
- b) Data Cleaning → Data Collection → Understanding the Problem → Data Analysis
- c) Data Collection → Data Analysis → Understanding the Problem → Data Cleaning

Section B: Short Questions

Answer each question in 2–5 sentences. No answers are provided.

Topic: Data and Data Types

1. What is data? Give three different forms that data can take.
 2. Write two differences between qualitative and quantitative data.
 3. What are the key characteristics of qualitative data? Give two examples.
 4. What are the key characteristics of quantitative data? Give two examples.
 5. Give three examples of qualitative data and three examples of quantitative data.
 6. What type of data is the number of students in your class? Justify your answer.
 7. What type of data is the favourite colour of students in a class? Justify your answer.
 8. Explain the "arithmetical" characteristic of quantitative data with an example.
 9. Explain the "measurable" characteristic of quantitative data with an example.
 10. Explain the "countable" characteristic of quantitative data with an example.
-

Topic: Organising and Analysing Data

1. Why is it important to organise data before analysing it? Give two reasons.
 2. Why is it important to organise data into tables or charts before analysing it?
 3. What is a data table? How does it help in data analysis?
 4. What are charts? State three common types of charts.
 5. What are graphs? State four common types of graphs.
 6. What is a bar chart used for? Give one example.
 7. For which purpose do we use a line graph?
 8. What is the difference between a bar chart and a pie chart?
 9. When would you use a scatter plot instead of a line graph?
 10. How does organising data reduce errors? Give an example.
-

Topic: Data Collection Methods and Tools

1. What is data collection? Why is it important?
 2. What is a survey? How is it different from a questionnaire?
 3. Which method would you use to collect opinions from a large group of people about a new school policy? Justify your answer.
 4. What is an interview as a data collection method? When is it most useful?
 5. What is observation as a data collection method? Give one example.
 6. What are online data sources? Give two examples of situations where you would use them.
 7. List four best practices in survey design and administration.
 8. Name three online tools that can be used to create surveys and briefly describe each.
 9. Differentiate between surveys and interviews as data collection methods.
 10. You want to find out the most popular sport among students in your school. Which data collection method would you choose and why?
-

Topic: Structured and Unstructured Data

1. What is structured data? Give two examples.
 2. What is unstructured data? Give two examples.
 3. Differentiate between structured and unstructured data.
 4. Give one example of structured data and explain why it is structured.
 5. Give one example of unstructured data and explain why it is unstructured.
 6. Why is structured data easier to search and analyse than unstructured data?
-

Topic: Data Visualisation

1. What is data visualisation? Why is it important?
 2. Explain why data visualisation is important in data analysis.
 3. Name four popular data visualisation tools and briefly describe what each one does.
 4. What is Microsoft Excel used for in data visualisation?
 5. How does Microsoft Power BI help in data visualisation?
 6. What makes Tableau different from Excel and Google Sheets as a visualisation tool?
 7. How does turning data into a chart or graph help in drawing conclusions?
-

Topic: Data Pre-Processing

1. What is data pre-processing? Why is it the most important first step in working with data?
 2. What does evaluating data quality involve? What questions should you ask?
 3. What is an error in data? Give one example.
 4. What is an outlier in data? Give one example.
 5. What is a bias in data? Give one example.
 6. What is the difference between an error and an outlier in a dataset?
 7. What is data validation? State two types of data validation.
 8. What does validating data completeness mean? Give an example.
 9. What does validating data accuracy mean? Give an example.
 10. What is data cleaning? What three things does it involve?
 11. How do you handle missing data during data cleaning? Give one strategy.
 12. How do you deal with outliers during data cleaning?
 13. A student's score is recorded as 105 out of 100. What type of data quality issue is this and how would you fix it?
-

Topic: Data Analysis

1. What is data analysis? What is its purpose?
 2. What is quantitative analysis? What type of data does it deal with?
 3. What is qualitative analysis? What type of data does it deal with?
 4. Differentiate between quantitative analysis and qualitative analysis.
 5. Give one example of a situation where quantitative analysis would be used.
 6. Give one example of a situation where qualitative analysis would be used.
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Topic: Cloud Storage and Data Backups

1. What is cloud storage? What is its main advantage over traditional storage?
2. What is remote access? How is it useful for students and professionals?
3. What is a data backup? Why is it important?
4. Give one example of how a student can back up their school project.
5. List three events that could lead to data loss that backups can protect against.
6. What is version control in collaborative tools? Why is it useful?
7. State two benefits of using collaborative cloud-based tools in education.
8. How does cloud storage enable global collaboration? Give an example.

Topic: Introduction to Data Science

1. What is data science? Define it in your own words.
 2. Why is data science important in everyday life? Give two examples.
 3. What are the three fields that make up the interdisciplinary nature of data science? Briefly explain each one.
 4. How is data science used in sports? Give a specific example.
 5. How is data science used in healthcare? Give a specific example.
 6. How is data science used in business? Give a specific example.
 7. Explain the detective analogy used in the chapter to describe data science.
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Topic: Data Science Tools

1. Name four data science tools and state what each one is used for.
 2. How can we use Python in data science?
 3. What is the role of the Pandas library in Python?
 4. What is the role of the Matplotlib library in Python?
 5. What is R? What is it particularly good at?
 6. What is SQL? What is it used for?
 7. How does SQL help a data scientist extract specific information from a large database? Give an example.
 8. Differentiate between Python and R as data science tools.
 9. Differentiate between Excel and Tableau as data visualisation tools.
 10. Which data science tool would you use to manage and query a large student database? Justify your answer.
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Topic: Big Data

1. What is big data? What makes it different from ordinary data?
 2. Name four domains where big data is applied and give one example from each.
 3. What is predictive modelling in data science? Give one example.
 4. What is graph analytics? What is it used to study?
 5. What is data analytics? How is it related to data science?
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Section C: Long Questions

Answer each question in full detail. Use examples, diagrams, or tables where applicable. No answers are provided.

1. Define data and explain the two broad categories of data — qualitative and quantitative. For each category:

- Provide a clear definition
 - List all key characteristics with explanations
 - Give at least three real-world examples
 - Explain how they differ from each other
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2. Explain the importance of organising data. Describe how data can be organised using tables, charts, and graphs. For each method:

- Explain what it is
 - State when it is most useful
 - Give a real-world example
 - Name at least two specific types (for charts and graphs)
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3. Describe five methods of data collection in detail. For each method:

- Provide a clear definition
- Explain how it works
- Give a real-world example
- State one advantage and one disadvantage

Which method would be most suitable for collecting opinions of 200 students about a new school canteen menu? Justify your answer.

4. Differentiate between structured and unstructured data. In your answer:

- Define both types clearly
 - Give at least two examples of each
 - Explain why structured data is easier to analyse
 - Describe one real-world scenario where structured data is used and one where unstructured data is generated
 - State which type of data is more common in the modern world and why
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5. What is data visualisation? Explain its importance in data analysis. Describe four data visualisation tools — Microsoft Excel, Google Sheets, Tableau, and Microsoft Power BI — and compare their features and uses. Which tool would you recommend for a student working on a school project and why?

6. Explain data pre-processing in detail. In your answer:

- Define data pre-processing and explain why it is the first and most important step
 - Describe what evaluating data quality involves
 - Define and give examples of errors, outliers, and biases in data
 - Explain data validation — both completeness and accuracy — with examples
 - Explain data cleaning — including error correction, handling missing data, and dealing with outliers
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7. Describe the two main types of data analysis — quantitative and qualitative. For each:

- Provide a clear definition
- Explain what type of data it deals with
- Give two real-world examples of when it would be used
- State one advantage

Compare the two types of analysis and explain a situation where you might need to use both together.

8. Explain the role of cloud storage and data backups in modern data management. In your answer:

- Define cloud storage and explain how it differs from traditional storage
 - Explain what remote access is and why it is useful
 - Define a data backup and explain its importance
 - Describe two benefits of collaborative cloud-based tools
 - Give three real-world examples of how cloud storage and backups are used in education or business
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9. What is data science? Explain its definition, scope, and importance. In your answer:

- Define data science clearly
- Explain the interdisciplinary nature of data science — the role of computer science, mathematics/statistics, and business knowledge

- Give examples of how data science is applied in at least four real-world domains (e.g., sports, healthcare, education, business)
 - Explain why learning data science is important for students today
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10. Describe the four main data science tools — Microsoft Excel, Python, R, and SQL. For each tool:

- Provide a clear description
 - Explain what it is used for in data science
 - State any specific libraries or features it offers
 - Give a real-world example of how it is used
 - State one advantage compared to the other tools
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11. Explain the concept of big data, its characteristics, and its applications. In your answer:

- Define big data and explain what makes it different from ordinary data
 - Describe the key characteristics of big data
 - Explain its applications in at least four domains: retail, healthcare, finance, and transportation
 - Explain what predictive modelling is and give one example
 - Explain what graph analytics is and give one example
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12. Write a comprehensive essay on the complete data science process. Your essay should cover:

- What data is and the types of data (qualitative vs quantitative)
 - How data is collected (methods and tools)
 - How data is organised and visualised
 - How data is pre-processed and cleaned
 - How data is analysed (quantitative and qualitative techniques)
 - How data is stored and protected (cloud storage and backups)
 - The ethical considerations involved in working with data (privacy, confidentiality, legal frameworks)
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13. Discuss the ethical practices related to data in detail. In your answer:

- Explain the importance of data privacy
- Explain what data confidentiality means and why it matters
- Describe the legal frameworks that protect data
- Explain what secure data handling involves

- Give real-world examples of what can happen when ethical data practices are not followed
 - State three best practices that individuals and organisations should follow to handle data ethically
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End of Unit 6: Data Science and Data Gathering — Complete Question Bank